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Physical and mechanical characteristics of new basalt-FRP bars for reinforcing concrete structures

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HIGHLIGHTS

- Physical and mechanical characterization of newly developed basalt FRP bars for concrete members.
- Durability of BFRP bars in harsh alkaline conditions simulating the concrete environment.
- Discussion of BFRP characterization and durability results and compare them against the requirements of FRP standards.

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ABSTRACT

This paper introduces an experimental study that investigated the physical, mechanical, and durability characteristics of basalt fiber-reinforced polymer (BFRP) bars. Durability and long-term performance were assessed by conditioning the BFRP bars in an alkaline solution simulating the concrete environment (up to 3000 h at 60 °C) to determine their suitability as internal reinforcement for concrete elements. Thereafter, the properties were assessed and compared with the unconditioned (reference) values. In this study, three types of BFRP bars were investigated. The test results revealed that the BFRP bars had good mechanical behavior and could be placed in the same category as grade II and grade III GFRP bars (according to tensile modulus of elasticity). Their tensile strength, however, was higher than that provided by CAN/CSA S807-10 for CFRP bars. On the other hand, the BFRP bars showed poor alkali resistance and exhibited a remarkable reduction in mechanical properties due to the resin–fiber interface issues, which needs to be remedied to achieve the desired durability characteristics.

1. Introduction

Premature degradation of concrete structures due to the corrosion of embedded steel reinforcement bars is a well-known and well-documented problem, particularly where deicing salts are routinely used, such as on concrete deck slabs and in parking garages [1]. The use of fiber-reinforced polymer (FRP) composite materials, for concrete structures located in severe adverse environments, has achieved worldwide acceptance because of their noncorrodible nature, higher tensile strength, and lower weight relative to conventional steel reinforcing bars. FRPs are available with a wide range of mechanical properties (tensile strength, bond strength, and modulus of elasticity) and are made with high-tensile-strength fibers such as carbon, glass, and aramid embedded in polymer matrices such as vinylester and epoxy. Moreover, FRPs can be produced as bars, ropes, tendons, and grids in a wide variety of shapes and surface configurations as well as with varied characteristics.

In a continuous effort to develop FRP technology and innovate, new types of fibers—such as basalt fibers—are being introduced to manufacture basalt fiber-reinforced polymers (BFRPs). BFRP is the most recently FRP composite, appearing within the last decade. BFRP has good potential to provide benefits that are comparable or superior to other FRP types and provide significantly better cost-effectiveness compared to carbon FRP (CFRP). Basalt fibers show high tensile strength and modulus, good chemical resistance,

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extended operating-temperature range, and good environmental friendliness. Therefore, basalt fibers are ideally suited for applications involving high temperature, chemical resistance, durability, mechanical strength, and low water absorption [2]. In comparison to E-glass FRP, BFRP has higher strength and modulus, similar cost, and greater chemical stability. Moreover, it exhibits over five times the strength and around one-third the density of commonly used low-carbon steel bars [3]. Few studies, however, have investigated the performance of BFRP bars, which underscores the importance of assessing their physical, mechanical, and durability characteristics in order to understand their behavior and generate higher confidence in these newly developed materials.

2. Literature review

Basalt fiber is an inorganic fiber made from quarried basalt rock by melting the rocks at 1400 °C. The molten rocks are then extruded through small nozzles to produce continuous filaments of basalt fibers. The fibers typically have diameters ranging from 9 to 13 μm [4]. Basalt fibers have many excellent characteristics: energy-saving; environmentally friendly; natural green fiber; high tensile strength and modulus; a wide range of working temperatures (−269 to 700 °C); good acid, salt, and alkali resistance; anti-ultraviolet; low moisture absorption; good insulation; anti-radiation; and sound wave-transparent properties [3,5–8]. The chemical composition of basalt fibers closely resembles that of the commonly used E- and S-glass fibers, except that basalt contains a high ratio of iron, which yields its brown color, as shown in Table 1 [9]. Fig. 1 shows some basalt fibers. The mechanical properties of basalt fibers from different sources are also different [7], probably due to different chemical components and processing conditions, such as drawing temperature. The tensile strength of basalt fibers tends to increase with increasing drawing temperatures. This is due to increasing proportions of crystal basalt nuclei at lower temperatures, as proved by scanning electron microscopy (SEM) [10].

Table 1
Chemical composition comparison between basalt and glass FRP.

Chemical composition (%)	Basalt fibers	E-glass fibers	S-glass fibers
Silicon dioxide, SiO_2	48.8–51.0	52–56	64–66
Aluminum oxide, Al_2O_3	14.0–15.6	12–16	24–26
Iron oxide, $FeO + Fe_2O_3$	7.3–13.3	0.05–0.40	0–0.3
Calcium oxide, CaO	10.0	16–25	0–0.3
Magnesium oxide, MgO	6.2–16.0	0–5	9–11
Sodium oxide & potassium oxide, $Na_2O + K_2O$	1.9–2.2	0–2	0–0.3
Titanium oxide, TiO_2	0.9–1.6	0–0.8	–
MnO	0.10–0.16	–	–
Fluorides	–	0–1	–
Boron oxide	–	5–10	–

Wu et al. [11] assessed the residual tensile properties of unstressed and stressed BFRP bars exposed to four types of simulated harsh environments: alkaline solution, salt solution, acid solution, and deionized water at 25, 40, and 55 °C. Microstructural analysis with scanning electronic microscopy (SEM) was also performed to reveal the inherent degradation mechanism of BFRP bars in an alkaline environment. The residual tensile strength of unstressed BFRP bars exposed to an alkaline solution was used to predict long-term performance based on the Arrhenius theory. The results showed that the effect on the durability of BFRP bars exposed to acid, salt, and deionized water was less than that for bars exposed to alkaline solution. The effects of sustained stress on the degradation of BFRP bars were not obvious when the stress level was less than 20% of ultimate strength, but the degradation processes accelerated when the stress exceeded this level. Considering a mean annual temperature of 5.7 °C (which represents an area with a northern latitude of 50°), the predicted exposure time to produce a 50% reduction in strength was estimated at approximately 16.1 years for the 6 mm BFRP bar.

Sim et al. [7] investigated the durability and elevated-temperature performance of basalt, glass, and carbon fibers. They reported that when the fibers were immersed in an alkali solution, the basalt and glass fibers lost volume and strength (50% at 7 days and more than 80% at 28 days) with a reaction product on the surface, but the carbon fiber did not show significant strength reduction (about 13%). Nevertheless, the basalt fiber maintained its volumetric integrity and 90% of its strength after exposure to high-temperature at over 600 °C for 2 h. Wang et al. [12] studied the chemical durability and mechanical properties of basalt fiber and its epoxy resin composites. The experimental results showed that, after the BFRP was exposed to alkali solutions (including saturated Na_2CO_3 solution, 10% NaOH, and 10% $NH_3{\cdot}H_2O$) for 3 months, the modulus remained unaffected, although the strength decreased by 40%. Li et al. [13] studied the durability and fatigue performance of basalt-fiber/epoxy-FRP bars exposed to hygrothermal and alkaline environments. The bare basalt fibers (no resin protection) immersed in these environments exhibited a severe degradation of tensile properties due to significant fiber corrosion, as revealed by SEM. In contrast, the BFRP bars evidenced higher durability when subjected to the same conditions.

This paper presents an experimental investigation aimed at assessing the physical and mechanical characteristics of newly developed BFRP bars. It also aimed at assessing their long-term durability through conditioning in an alkaline solution simulating a moist concrete environment at high temperature. Since BFRPs have not been included in design standards and specifications yet, the results of this experimental study will contribute to integrating BFRP into FRP standards and guides.

Fig. 1. Basalt fibers.

3. Experimental program

This experimental work was carried out on newly developed BFRP bars to investigate their short- and long-term characteristics. This research work is a part of an extensive research project being conducted at the University of Sherbrooke through NSERC Research Chair activities to develop and introduce new FRP reinforcement for concrete infrastructure especially that subjected to harsh environmental conditions, such as marine structures, parking garages, and bridge deck slabs. The findings of this research project will contribute to integrating BFRP into FRP standards and guides, such as ACI 440.1R-06 [14], ACI 440.6M-08 [15], CAN/CSA S807-10 [16], CAN/CSA S6S1 [17], and CAN/CSA S806-12 [18].

The experimental program involved three types of BFRP bars: A, B, and C. Type A has a 7 mm diameter and is manufactured for prestressing purposes. Its physical and mechanical properties as well as durability in a concrete environment were investigated. Further study to investigate prestressing-related characteristics such as relaxation under sustained loads and creep–rupture will be conducted in another research project. Types B and C bars are 8 mm in diameter with similar surface configurations. They were provided by the manufacturer with an expected tensile modulus close to 50 and 60 GPa, simulating grade II and III GFRP bars [16]. The research program was divided into three phases. Phase I focused on physical characterization of the BFRP bars. The physical properties determined in this phase served as references for physical properties after conditioning. Phase II focused on mechanical characterization of the BFRP bars. The tensile strength, tensile modulus of elasticity, ultimate tensile strain, transverse-shear strength, flexural strength, flexural modulus of elasticity, interlaminar-shear strength, and bond strength were determined according to the appropriate test methods. The test results also served as references for calculating the residual strengths after conditioning. Phase III assessed the durability and long-term performance of the conditioned BFRP bars. The durability was assessed by immersing the BFRP specimens in an alkaline solution at high temperature (60 °C) for different lengths of time (3000 h for Type A and 2160 h for Types B and C) designed to simulate a concrete environment so as to validate the performance of the BFRP bars as internal reinforcement for concrete elements. Changes in the physical and mechanical characteristics were assessed by comparing the characteristics of the conditioned BFRP bars to the reference ones from Phases I and II.

4. Materials and test procedures

Three types of BFRP bars were used in this study. Type A has a 7 mm diameter (nominal cross-sectional area of 38.46 mm^2) with a woven surface, while Types B and C have a 8 mm diameter (nominal cross-sectional area of 50.24 mm^2) with a deformed surface. The BFRP bars were made of continuous basalt fibers impregnated in vinylester resin according to the pultrusion process. Fig. 2 shows the BFRP bars. The basalt fibers used herein were known ASA.TEC (produced by Asamer Basaltic Fibers GmbH, Austria). The fibers were produced from volcanic material with organic surface coating and had a diameter of 10–19 μm.

The tests in Phase I (physical characterization) were conducted in accordance with ACI [15] and CSA [16] and the relevant ASTM standards. The relative density was determined according to ASTM D792 [19], fiber content according to ASTM D3171 [20], transverse coefficient of thermal expansion according to ASTM E831 [21], water absorption according to ASTM D570 [22], cure ratio according to ASTM D5028 [23], and glass-transition temperature (T_g) according to ASTM D3418 [24]. In addition, microstructural analysis was performed for all three types of BFRP specimens using scanning electron microscopy (SEM) for both the unconditioned (reference) and conditioned specimens to assess changes and/or degradation. The effects of conditioning on the glass-transition temperatures (T_g) and on BFRP-bar chemical composition were also determined with differential scanning calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively.

The Phase II mechanical-characterization tests were tensile strength (ASTM D7205 [25]), transverse-shear testing (ASTM D7617 [26]), flexural testing (ASTM D4476 [27]), short-beam shear testing (ASTM D4475 [28]), and bond strength using the pullout test (ACI 440.3R-12 [29], B.3 Test Method). The mechanical properties reported herein were calculated using the nominal cross-sectional areas.

Furthermore, a pilot investigation was also conducted to evaluate the effects of chemicals on the basalt fibers used in manufacturing the BFRP bars tested herein. The tests were conducted according to Owens Corning [30] guide for glass fibers and the results are reported.

4.1. Chemical resistance evaluation of bare basalt fibers

Chemical resistance tests were conducted on the basalt fibers utilized in manufacturing the tested BFRP bars, in parallel to the main study conducted herein to investigate the physical and mechanical characteristics of new BFRP bars. These tests were conducted to clarify the effect of the different chemicals on the bare basalt fibers. The tests were conducted following Owens Corning [30] guide for evaluating the chemical resistance of glass fibers. Basalt fibers were heated to 540 °C overnight to remove any sizing and provide a proper clean surface for chemical-resistance investigation. Samples were then cut and carefully weighed before immersion in different corrosive aqueous solutions at 96 °C for 1 (24 h) and 7 days (168 h). The solutions were deionized water, an acidic solution (10% HCl), a saline solution (10% NaCl), and an alkaline solution (3.2 g NaOH per liter). After conditioning, the samples were thoroughly washed, dried, and weighed again. Table 2 presents the mass losses after 1 and 7 days of conditioning in deionized water, acidic solution, saline solution, and alkaline solution. The mass losses were less than 1% (wt) after immersion in water and saline solution, whereas the alkaline solution caused a mass loss of about 3% (wt) and the acidic solution caused a mass loss of 7.0% (wt). It should be mentioned that the color of the acidic solution turned green after conditioning, which may indicate some leaching of iron oxide.

Furthermore, the conditioned fibers were then analyzed using SEM to detect any changes in microstructure. Fig. 3 presents

Table 2
Mass loss (wt%) of basalt fibers after conditioning in water, acidic, saline, and alkaline solutions at 96 °C.

Conditioning period (days)	Mass loss (wt.%)			
	Water	Acidic	Saline	Alkaline
1	0.3	5.9	−0.6	0.7
7	0.6	7.0	0.2	2.6

Type A BFRP   Types B and C BFRP

Fig. 2. Tested BFRP bars.

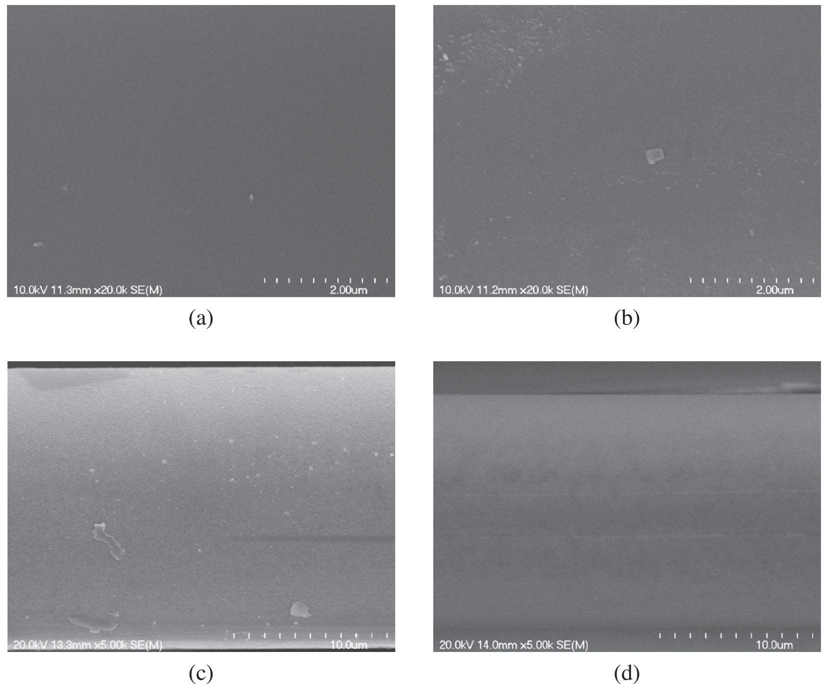


Fig. 3. SEM micrographs of basalt fibers. (a) Before condition (reference); (b) after conditioning in acidic solution; (c) after conditioning in saline solution; and (d) after conditioning in alkaline solution.

typical micrographs obtained on unconditioned (reference) and conditioned samples (acidic, saline, and alkaline solutions). The basalt fiber conditioned in the different solutions showed tiny signs of corrosion.

5. Testing, results, and discussion

5.1. Phase I: physical characterization

The physical properties of the reference (unconditioned) BFRP bars were determined according to ACI [15] and CSA [16] test methods. Since BFRP bars are not included in any FRP standard yet, the physical properties of the investigated BFRP bars and tendons were compared to the specified limits for FRP bars. Table 3 presents the results of those physical characterization tests and compares them to the limits specified in ACI [15] and CSA [16] for FRP bars. Table 3 shows that the fiber content of the BFRP bars was 85.1%, 77.4%, and 81.1% for BFRP Types A, B, and C, respectively, thereby satisfying the ACI limits [15] (55% by volume) and CSA limits [16] (70% by weight). The cure ratio of Type A BFRP was 97.5%, while the curing ratio of Types B and C was 100%. The BFRP specimens had a transverse coefficient of thermal expansion (CTE) ranging from 18.4×10^{-6} to 26.8×10^{-6} °C, which is less than 40×10^{-6} °C as stated by CSA [16]. Type B's higher transverse CTE as compared to Types A and C may be due to the lower fiber content (higher resin content).

Differential scanning calorimetry (DSC) is used to obtain information about the thermal behavior and characteristics of polymer materials and composites, such as the glass-transition temperature (T_g) and curing process. In this study, 30–50 mg specimens from both the unconditioned and conditioned specimens were sealed in aluminum pans and heated in a TA Instruments DSC Q10 calorimeter to 200 °C at a rate of 20 °C/min. The glass-transition temperature (T_g) was determined in accordance with ASTM D3418 [24]. Two scans were performed for each BFRP type. Fig. 4 presents the DSC scans for the T_g, while Table 3 presents the T_g values. Since the cure ratio of BFRP type A was 97.5%, this indicates the presence of uncured resin. Consequently, BFRP type A showed a T_g of 105 °C in the first run and 123 °C in the second run because of the post-polymerization of uncured resin. BFRP types B and C evidenced a T_g of 119 and 127 °C, respectively, with no post-polymerization, since the T_g in the second run was almost

Table 3
Results for physical properties of BFRP bars and the specified limits for FRPs.

Property	Type (A)	Type (B)	Type (C)	Specified limits for FRP bars	
				ACI [15]	CSA [16]
Relative density (ρ)	2.167	1.998	2.046	–	–
Fiber content by weight (%)	85.1	77.4	81.1	55% (by volume)	70 (by weight)
Transverse CTE ($\times 10^{-6}$ °C^{-1})	18.7	26.8	18.4	NA	40
Cure ratio (%)	97.5	100	100	NA	⩾93 (D2*); 95 (D1*)
T_g (°C)					
Run 1	105	118	127	100	⩾80 (D2*); 100 (D1*)
Run 2	123	119	129		
Moisture uptake (%)	0.15	0.56	0.62	1.0	1.0 (D2*); 0.75 (D1*)

* Classification based on durability: FRPs with high durability shall be classified as D1; FRPs with moderate durability shall be classified as D2; FRPs made with vinylester and epoxy shall be classified as D1 or D2; FRPs made with polyester matrix shall be classified as D2.

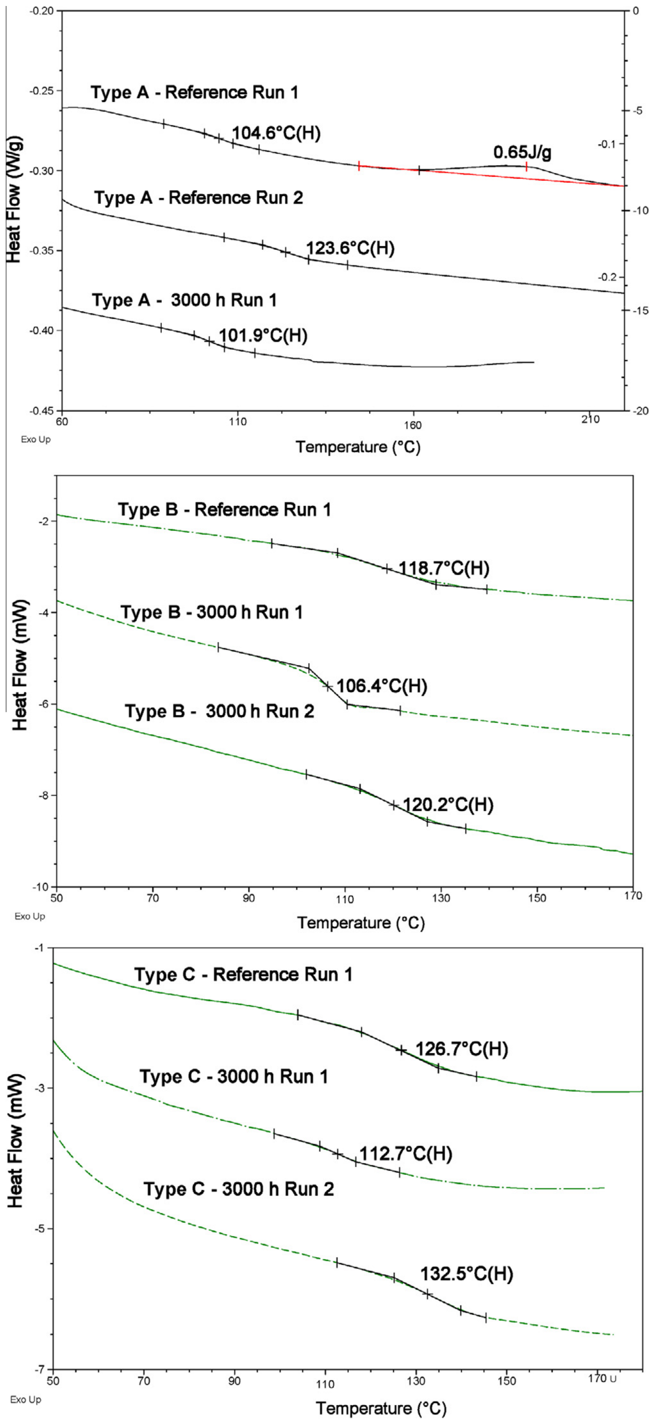


Fig. 4. DSC graphs for glass-transition temperature (T_g).

the same as in the first. It should be mentioned that the higher T_g of the specimens after the second scan is an indicative of a reversible chemical reaction and consequently not caused by any chemical degradation of the resin (such as hydrolysis).

The moisture uptake at saturation was determined. The test was carried out in accordance with ASTM D570 [22]. Five 100 mm long specimens for each type of BFRP were cut, dried, and weighed. They were then immersed in water at 50 °C for 3 weeks. The samples were periodically removed from the water, surface dried, and weighed. The water content as a percentage of weight was calculated with Eq. (1).

$$W = \frac{(P_s - P_d)}{P_d} \times 100 \tag{1}$$

where P_s and P_d are the bar weights in the saturated and dried states, respectively. The percentage of moisture uptake was calculated. The gain in mass was corrected to account for specimen mass loss due to a possible dissolution phenomenon. This correction was achieved by completely drying the immersed specimens in an oven at 100 °C for 24 h and comparing their masses to their initial masses. The moisture uptake was 0.15%, 0.56%, and 0.62% for types A, B, and C, respectively. This difference is due to delamination and/or debonding of the fiber–resin interface or due to the presence of continuous voids, which will be discussed later in this paper.

It should be mentioned that the BFRP bars met the requirements of D1 FRP bars (FRP bars with high durability [16]) with respect to cure ratio, T_g, and moisture uptake. Regardless of the differences between the results for the three BFRP products tested, they still meet the physical-property requirements of ACI [15] and CSA [16].

5.2. Phase II: mechanical characterization

The mechanical characterization included testing of representative BFRP specimens to determine their tensile properties in accordance with ASTM D7205 [25], transverse-shear strength in accordance with ASTM D7617 [26], flexural properties in accordance with ASTM D4476 [27], interlaminate shear strength in accordance with ASTM D4475 [28], and bond strength in accordance with ACI 440.3R-12 [29] B.3 Test Method and CSA [18], Annex G. Figs. 5–9 show the mechanical characterization tests and the tested specimens at failure, while Table 4 lists the results. The following sections provide brief descriptions and interpretation of those results.

5.2.1. Tensile properties of the reference BFRP bars

The tensile properties of FRP bars are among the most important parameters affecting the design of FRP-reinforced concrete sections, since tensile strength governs section capacity and the tensile modulus of elasticity governs the serviceability limit state. The tensile properties of the investigated BFRP bars were determined by testing five representative specimens for each type in accordance with ASTM D7205 [25]. The specimens were cut to the desired lengths and prepared by installing the steel tube (anchors) with expansive cement grout commercially known as Bustar Expanding Grout. The specimens were instrumented with a 200 mm LVDT to capture specimen elongation during testing. The tests were conducted with a Baldwin testing machine in which the applied load and specimen elongation were electronically recorded during the test. Fig. 5(a) shows the test setup. The tensile strength and tensile modulus of the BFRP bars were determined with Eqs. (2) and (3), respectively,

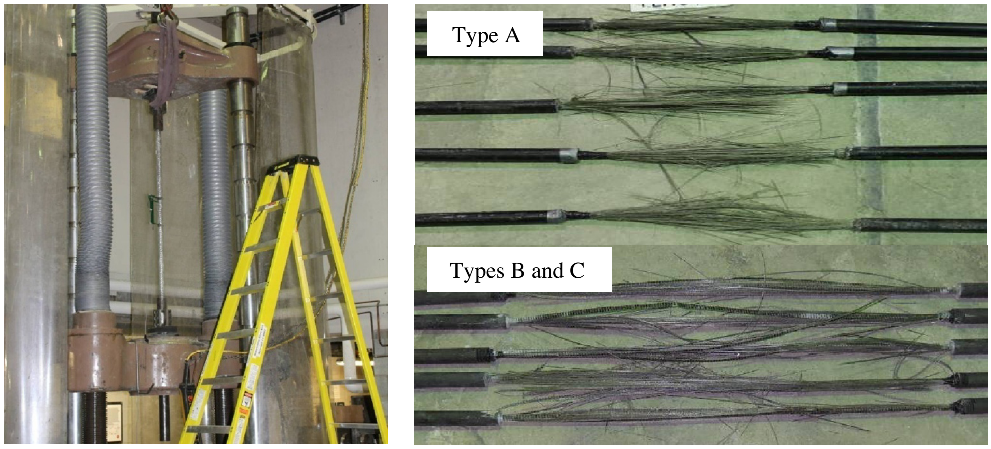
$$f_u = F_u/A \tag{2}$$

where f_u is the tensile strength (MPa), F_u is the tensile capacity (N), and A is the bar's cross-sectional area (mm^2).

$$E = (F_1 - F_2)/((\varepsilon_1 - \varepsilon_2)A) \tag{3}$$

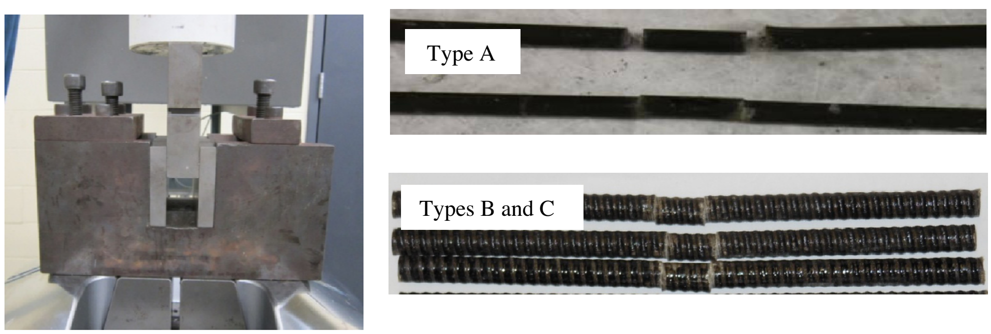
where E is the tensile modulus of elasticity (MPa); A is the cross-sectional area (mm^2); F_1 and ε_1 are the load and corresponding strain, respectively, at approximately 50% of the ultimate tensile capacity; and F_2 and ε_2 are the load and corresponding strain, respectively, at approximately 25% of the ultimate tensile capacity. The tensile properties of the tested specimens are listed in Table 4 and the failure is shown in Fig. 5(b).

Like all FRP products, the tested specimens showed a linear elastic stress–strain relationship up to failure. The tested bars



(a) Typical test setup (b) Failure mode (rupture)

Fig. 5. Typical tensile-strength test and failure mode.



(a) Test setup (b) Specimens at failure

Fig. 6. Typical transverse-shear strength and specimens at failure.

showed very high tensile strength exceeding 1500 MPa, which meets the requirements of GFRP (750 MPa for 8 mm diameter) and CFRP (1350 MPa for 8 mm diameter) bars according to CSA [16]. On the other hand, Types A and B showed tensile moduli of elasticity of 69.0 and 64.6 GPa, respectively, which correspond to the CSA [16] grade III (GIII) classification for GFRP bars. Type C, however, had tensile modulus of elasticity of 59.5 GPa, which corresponds to the CSA [16] grade II (GII) classification for GFRP bars. Furthermore, the tested specimens showed a strain at failure ranging from 2.43% to 2.64%, both higher than the 1.2% provided by CSA [16] for GFRP bars.

It should be mentioned that the design of FRP-reinforced members is based on guaranteed tensile strength (the mean tensile strength minus three times the standard deviation [14]).

5.2.2. Transverse-shear strength of the reference BFRP bars

Transverse-shear tests were conducted according to ASTM D7617 [26]. The setup consisted of a 230 × 100 × 110 mm steel base equipped with lower blades spaced at 50 mm face to face, allowing for the double transverse-shear failure of the specimen caused by an upper blade, as shown in Fig. 6(a).

For each type of BFRP bar, five unconditioned specimens of 200 mm in length were tested under laboratory conditions with an MTS 810 testing machine equipped with a 500 kN load cell. A displacement-controlled rate of 1.3 mm/min was used during the test, which yielded between 30 and 60 MPa/min until specimen failure. The loading was performed without subjecting the test specimens to any shock. Fig. 6(b) shows the specimens at failure. The transverse-shear strength was calculated with Eq. (4):

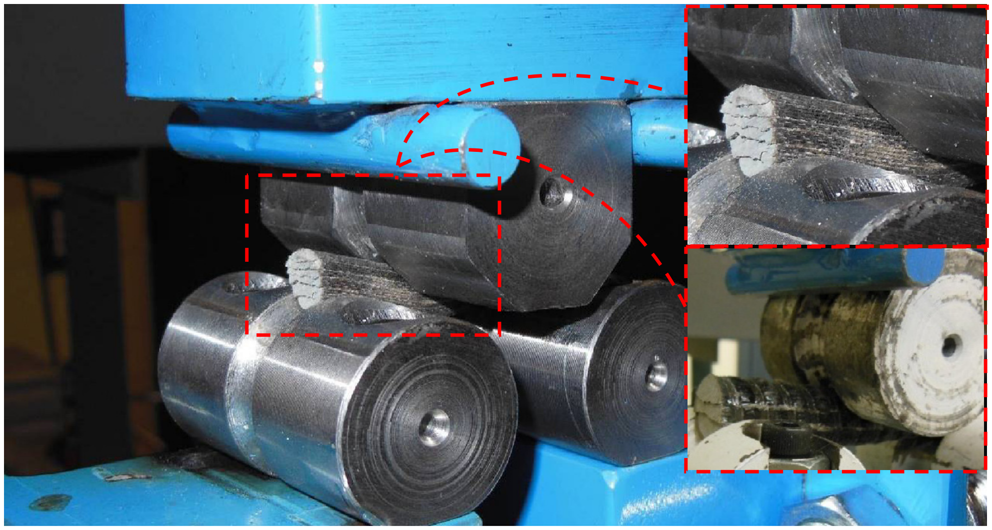
$$\tau_u = P_s/(2A) \tag{4}$$

where τ_u is the transverse-shear strength (MPa), P_s is the failure load (N), and A is the cross-sectional area of the FRP bar (mm^2).

As shown in Table 4, the transverse-shear strength of the BFRP specimens ranged from 293 to 344 MPa, with an average of 317 MPa. These values meet the requirements of CSA [16], which specifies a minimum transverse-shear strength of 160 MPa for FRP bars. These results were expected, since the transverse-shear strength resulted mainly from the resin with a small contribution from the fiber–resin interface [31].

5.2.3. Interlaminar-shear strength of the reference BFRP bars (short-beam shear test)

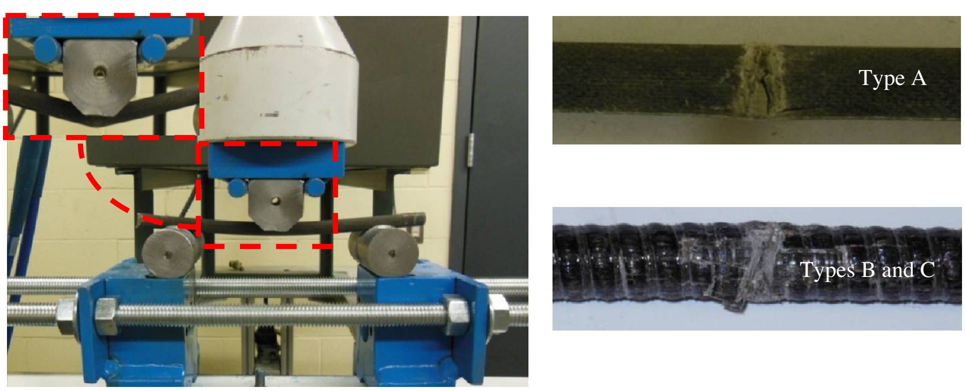
In pultruded FRP bars in which the fibers are arranged unidirectionally and bonded with the polymer matrix, the horizontal stresses would be more conducive to inducing interface degradation than transverse-shear stresses [32]. The short-beam shear test was conducted on five specimens of each type of FRP bars according to ASTM D4475 [28] in order to calculate the interlaminar-shear strength, which is governed by the fiber–matrix interface. The tests were carried out with a 500-kN MTS 810 testing machine. The distance between the shear planes was set to 6 times the nominal diameter of the FRP bars. Fig. 7 shows the test setup and typical mode of failure of the tested specimens. A displacement-controlled rate of 1.3 mm/min was used during the test. The applied load was recorded with a computer-monitored data-acquisition system.



(a) Test setup for the short-beam shear test

(b) Specimens at failure

Fig. 7. Interlaminar-shear-strength test (short-beam test) and specimens at failure.



(a) Test setup

(b) Specimens at failure (compression failure)

Fig. 8. Typical flexural-strength test and specimens at failure.

The interlaminar-shear strength, S_u, of the FRP bars was calculated with Eq. (5).

$$S_u = 0.849P/d^2 \quad (5)$$

where S_u is the interlaminar-shear strength (MPa), P is the shear failure load (N), and d is the bar diameter (mm).

Table 4 shows the interlaminar-shear strength of BFRP bars. BFRP Type B exhibited the highest interlaminar-shear strength (72 MPa), followed by Type A (63 MPa) and Type C (60 MPa). It is worth mentioning that the high values of the interlaminar-shear strength indicate a sound interface between the resins and reinforcing fibers. The difference between the highest (Type B) and the lowest (Type C) interlaminar-shear strength was 20%. This difference may have been affected by fiber–resin interface issue, which will be clarified in the SEM analysis.

5.2.4. Flexural properties of the reference BFRP bars (three-point flexural test)

Flexural testing is especially useful for quality control and specification purposes. Flexural properties may vary with specimen diameter, temperature, weather conditions, and differences in straining rates. The flexural properties obtained according to ASTM D4476 [27] cannot be used for design purposes but are appropriate for the comparative testing of composite materials.

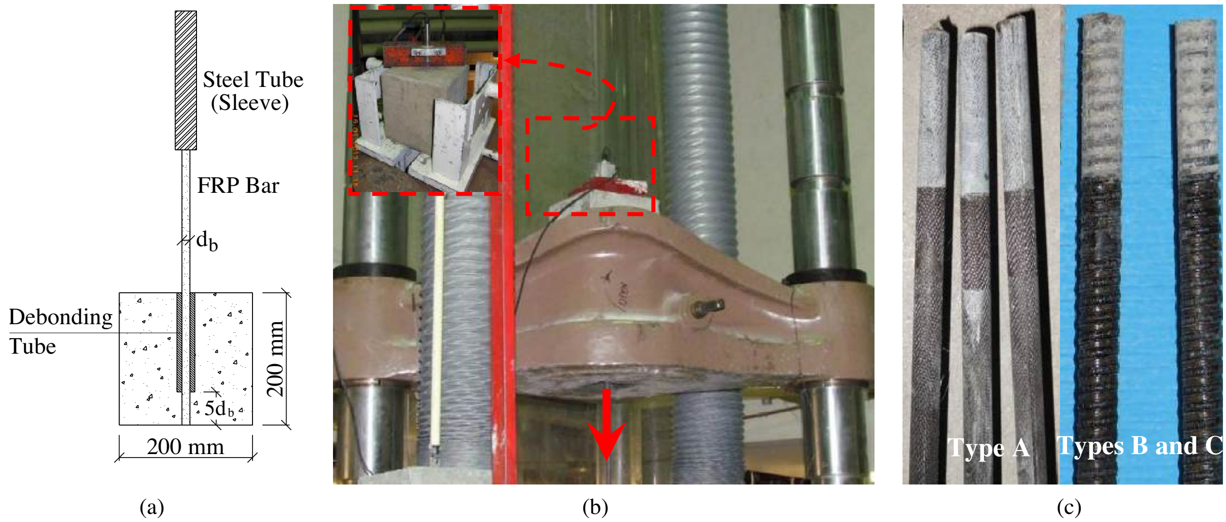


Fig. 9. Pullout test. (a) Specimen geometry; (b) test setup; (c) typical pullout bond failure.

Table 4
Mechanical properties of tested BFRP bars (reference/unconditioned).

BFRP type	Tensile properties [25]			Transverse-shear strength [26]	Interlaminar-shear strength[27]	Flexural properties [28]			Bond strength ACI [29]; CSA [18]
	f_u (MPa)	E (GPa)	ε_u (%)	τ_u (MPa)	S_u (MPa)	f_u (MPa)	E (GPa)	ε_u (%)	f_b (MPa)
A	1680 ± 133	69.0 ± 0.7	2.43	344 ± 18	63 ± 2.7	1790 ± 91	90.4 ± 3.8	1.98	7.0 ± 0.9
B	1655 ± 95	64.6 ± 1.4	2.56	315 ± 38	72 ± 2.9	2061 ± 216	84.6 ± 8.6	2.47	25.4 ± 2.3
C	1567 ± 115	59.5 ± 3.3	2.64	293 ± 28	60 ± 2.4	1587 ± 111	74.0 ± 4.2	2.15	27.2 ± 2.4

The test was conducted on specimens of 180 mm long over a simply supported span equal to 20 times the bar diameter, as shown in Fig. 8(a). Five unconditioned specimens were tested under laboratory conditions on an MTS 810 testing machine equipped with a 500 kN load cell as references for each type. The specimens were loaded at the mid-span with a circular nose; the specimen ends rested on two circular supports that allowed the specimens to bend. A displacement-controlled rate of 3.0 mm/min was used during the test. The rate of loading occurred without subjecting the test specimen to any shock. The applied load and deflection were recorded during the test on a computer- monitored data-acquisition system.

The flexural strength of tested FRP specimens was calculated with Eq. (6). Flexural modulus of elasticity is the ratio, within elastic limit, of stress to corresponding strain. It was calculated with Eq. (7):

$$f_u = PLC/(4I) \tag{6}$$

$$E = PL^3/(48IY) \tag{7}$$

where f_u is the flexural strength in the outer fibers at mid-span (N/mm^2), P is the failure load (N), L is the clear span (mm), I is the moment of inertia (mm^4), C is the distance from the centroid to the extremities (mm), E is the flexural modulus of elasticity in bending (N/mm^2), and Y is the mid-span deflection at load P (mm).

The maximum outer-fiber strain (ε_u) was calculated from Eq. (8).

$$\varepsilon_u = f_u/E \tag{8}$$

Table 4 provides test results of three-point loading test for the BFRP specimens in terms of flexural strength, flexural modulus of elasticity, and ultimate strain. The elastic behavior of all the specimens was retained until flexural failure, which occurred due to compression at the top fibers, as shown in Fig. 8(b). The moduli of elasticity from the flexural test were 24% to 31% higher than those from the tensile test. Since the ASTM D4476 [27] test method is designated for FRP bars with a diameter of at least 0.5 in. (12.7 mm), this may have impacted the flexural testing results.

5.2.5. Bond strength of the reference BFRP bars (pullout test)

The bond strength of the BFRP bars was assessed with the pullout test. The pullout tests were carried out in normal-strength concrete with a designated strength of 35 MPa after 28 days according to the ACI [29], B.3 Test Method and CSA [18], Annex G. The bonded length was kept constant at $5d_b$, where d_b is the nominal diameter of FRP bars. The pullout blocks measured 200 × 200 × 200 mm. Fig. 9 shows the geometry of the pullout specimens, test setup, and modes of failure, while Fig. 10 shows the typical bond

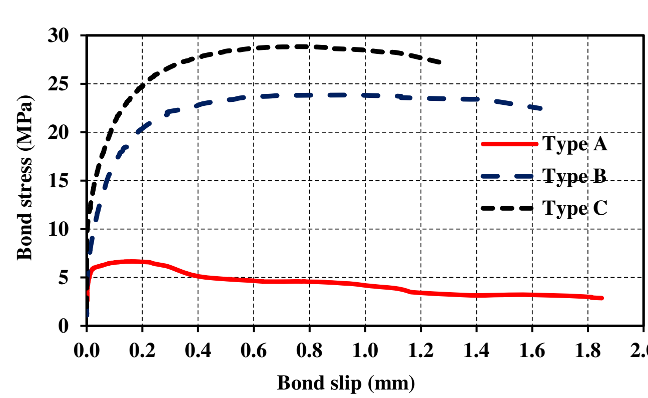


Fig. 10. Bond stress-free slip relationships of the tested BFRP bars.

stress-free slip relationship of the tested BFRP bars. Table 4 presents the bond strength of the tested BFRP bars.

The BFRP Type A (manufactured for prestressing purposes) showed an average bond strength of 7.0 MPa. According to CEB-FIP [33], the bond strength of prestressing steel bars of a diameter smaller than 32 mm in normal-weight concrete with compressive strength of 35 MPa is about 4 MPa for deformed bars and 2.0 MPa for smooth bars. Thus, the bond strength obtained herein is in agreement with what has been reported for steel tendons. On the other hand, the pullout tests of the Types B and C (with similar surface configurations) showed an average bond strength of 24.5 and 27.2 MPa, respectively, which is higher than 8.0 MPa as specified by CSA [16]. Thus, it could be concluded that Types B and C meet the mechanical-property requirements of CSA [16].

5.3. Phase III: durability study and long-term performance assessment

5.3.1. Conditioning of the BFRP specimens

This phase was conducted to assess the durability and long-term performance of the BFRP bars conditioned in an alkaline solution simulating concrete pore solution. Accelerated aging tests were conducted in accordance with ASTM D7705 [34]. The conditioning of the FRP bars included a combined exposure to an alkaline environment and elevated temperature. Immersion in an aqueous media (alkaline solution) at high temperature accelerates degradation. The alkaline solution was prepared to have a composition representative of the pore water inside Portland-cement concrete, specifically, 118.5 g of $Ca(OH)_2$, 0.9 g of NaOH, and 4.2 g of KOH per liter of deionized water. The solution had a pH of 12.6–13.0, which is representative of a mature concrete pore solution. The three types of BFRP bars were immersed in this solution at 60 °C. The BFRP specimens were conditioned for different lengths of time (1000 and 3000 h for BFRP Type A; 720 and 2160 h for BFRP Types B and C). The conditioning time started once the solution had reached the prescribed temperature. Robert et al. [35] reported that the increase in degradation reaction rate was almost linear between room temperature and 50 °C, whereas, at higher temperatures (over 60 °C), the increase in the degradation reaction rate was exponential. Therefore, to avoid any thermal degradation, the maximum conditioning temperature used in this study was 60 °C, as specified in ASTM D7705 [34].

The BFRP specimens were placed in hermetically sealed stainless-steel containers to prevent excessive evaporation and reaction of atmospheric CO_2 with calcium hydroxide. The containers were placed in an environmental chamber adjusted to the prescribed temperature (60 °C) under isothermal conditions. The BFRP bars were weighed and their diameters measured throughout the conditioning period to monitor water absorption and eventually characterize the mass and diameter changes. Observation revealed no changes in diameter during the conditioning periods. Specimens of each type of BFRP bar were removed from the solution and tested to determine their physical and mechanical properties after the exposure periods at 60 °C. Similar to as in Phase I, the effects of conditioning on the glass-transition temperatures (T_g) and chemical composition were also assessed with differential scanning calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively. The microstructure of all types of the BFRP bars was investigated with scanning electron microscopy (SEM) for both conditioned and unconditioned cases to assess changes and/or degradation. In addition, the mechanical properties of the conditioned specimens were assessed with tests similar to those in Phase II for tensile strength (ASTM D7205 [25]), transverse shear (ASTM D7617 [26]), flexural strength (ASTM D4476 [27]), short-beam shear (ASTM D4475 [28]), and bond strength using the pullout test (ACI [29], B.3 Test Method). The results for the conditioned specimens were compared to those of the reference ones. The change in the properties was selected as an indicator on the degradation of the BFRP materials.

5.3.2. Tensile properties of the conditioned BFRP bars

Table 5 presents the ultimate tensile strength, tensile modulus of elasticity, and ultimate tensile strain of the conditioned specimens along with the retention ratios. The results indicate that the BFRP Type A bars were highly affected by accelerated aging with an ultimate tensile strength retention of 60.9% after 3000 h in an alkaline solution at 60 °C. Types B and C were less affected by accelerated aging with ultimate tensile strength retentions of 77.0% and 76.5%, respectively, after 2160 h. Thus, BFRP Types B and C satisfied the 70% retention in CSA [16] for D2 bars, but these bars did not qualify for the 80% specified for D1 bars. On the other hand, as expected, the changes in the tensile moduli of elasticity were not significant for the tested BFRP specimens (less than 7.0%). Fig. 11(a) and (b) show the effect of the alkaline solution at high temperature (60 °C) on the tensile strength and modulus of elasticity of the BFRP specimens, respectively.

5.3.3. Transverse-shear strength of the conditioned BFRP bars

Table 6 shows the ultimate transverse-shear strength and the retention strength of the BFRP bars after conditioning. The Type A bars were affected by accelerated aging and had a transverse-shear strength retention of 81.7% after 3000 h. The Type B and C bars were slightly affected by accelerated aging and evidenced strength retentions of 89.6% and 87.0%, respectively, after conditioning for 2160 h. Fig. 11(c) shows the effect of the alkaline solution at high temperature (60 °C) on the transverse-shear strength of BFRP specimens.

5.3.4. Interlaminar-shear strength of the conditioned BFRP bars

Table 6 shows the interlaminar-shear strength of the conditioned BFRP bars and their retention-strength ratios. Fig. 11(d) shows the effect of the alkaline solution at high temperature (60 °C) on interlaminar-shear strength. Clearly, Type B has the highest interlaminar-shear strength and the lowest rate of degradation due to conditioning, followed by Types A and C. This observation coincides with the results for the reference specimens where Type B had an interlaminar-shear strength of 72 MPa compared to 63 and 60 MPa for Types A and C, respectively. This indicates that the Type B bars had a better fiber–matrix interface,

Table 5
Tensile properties and retention of the conditioned BFRP bars.

BFRP type	Cond. period (h)	f_u (MPa)	Retention (%)	E (GPa)	Retention (%)	ε_u (%)
A	1000	1012 ± 47.3	60.2	70.8 ± 1.2	102.6	1.43 ± 0.05
	3000	1023 ± 177.0	60.9	69.1 ± 0.4	100.0	1.48 ± 0.26
B	720	1429 ± 52.6	86.3	60.1 ± 0.7	93.0	2.38 ± 0.06
	2160	1275 ± 4.0	77.0	61.5 ± 0.9	95.2	2.07 ± 0.03
C	720	1409 ± 91.7	89.9	59.7 ± 1.6	100.0	2.36 ± 0.13
	2160	1198 ± 31.7	76.5	56.6 ± 0.9	95.1	2.12 ± 0.02

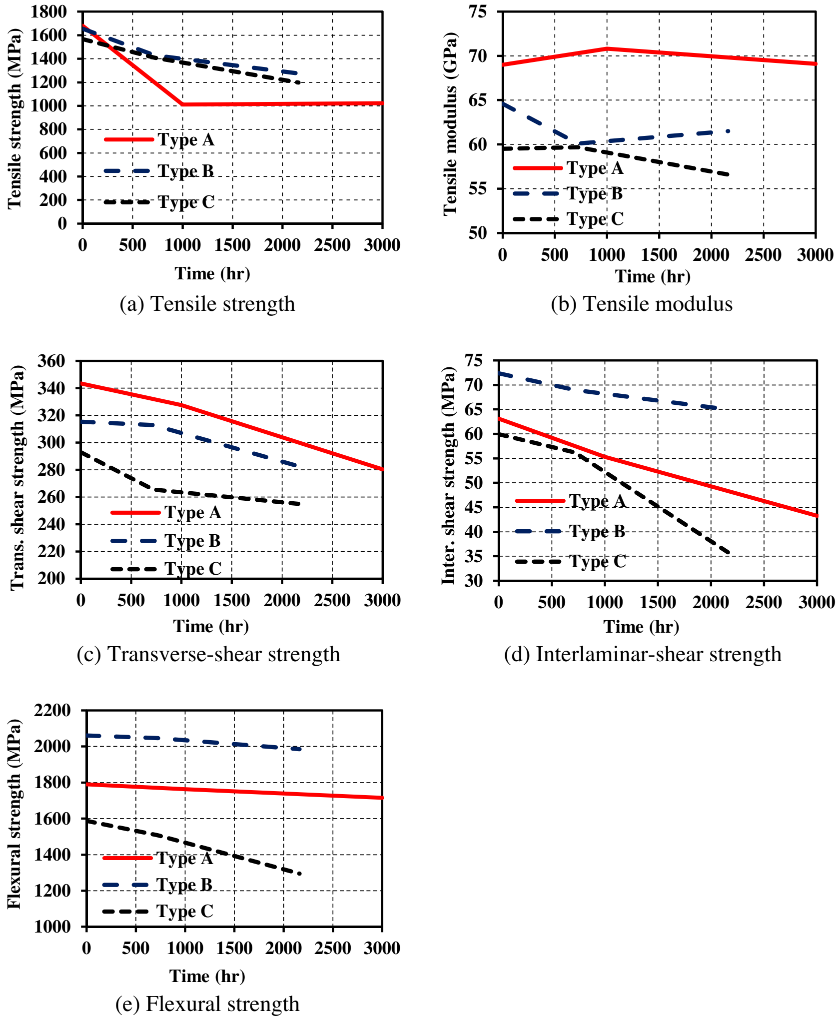


Fig. 11. Effect of conditioning on the mechanical properties of the tested BFRP bars.

Table 6
Transverse- and interlaminar-shear strength and retention of the conditioned BFRP bars.

BFRP type	Cond. period (h)	τ_u (MPa)	Retention (%)	S_u (MPa)	Retention (%)
A	1000	327.5 ± 7.6	95.4	55.3 ± 2.8	87.7
	3000	280.4 ± 7.3	81.7	43.25 ± 1.3	68.5
B	720	312.9 ± 36.6	99.2	68.98 ± 3.0	95.3
	2160	282.7 ± 30.0	89.6	65.00 ± 5.1	89.8
C	720	265.6 ± 11.3	90.6	56.20 ± 2.7	93.8
	2160	254.9 ± 15.9	87.0	35.82 ± 7.0	59.8

Table 7
Flexural properties and retention of the conditioned BFRP rods.

BFRP type	Cond. period (h)	f_u (MPa)	Retention (%)	E (GPa)	Retention (%)
A	1000	1763.3 ± 60.9	98.5	93.0 ± 2.4	102.8
	3000	1715.2 ± 100.7	95.6	90.8 ± 1.0	100.0
B	720	2047.0 ± 197.1	99.3	72.7 ± 2.7	85.9
	2160	1985.2 ± 77.9	96.3	66.7 ± 4.4	78.8
C	720	1507.4 ± 156.9	95.0	63.7 ± 1.3	86.1
	2160	1295.5 ± 153.5	81.6	63.0 ± 1.7	85.1

which minimized the degradation due to conditioning. The fiber–matrix interface was investigated, as described later, with SEM.

5.3.5. Flexural properties of the conditioned BFRP bars

Table 7 shows the flexural properties of conditioned BFRP bars as well as their retention strengths and modulus ratios after conditioning. Fig. 11(e) shows the effect of the alkaline solution on flexural strength. Type B bars have the lowest degradation rate, followed by Type A bars and Type C bars, the latter being significantly affected by the conditioning. This observation confirms good bonding between the basalt fibers and resin (fiber–matrix interface) in Type B, followed by Types A and C, which minimized degradation due to conditioning.

5.3.6. Glass-transition temperature of the conditioned BFRP bars

Differential scanning calorimetry (DSC) was also conducted on the specimens conditioned in the alkaline solution at 60 °C for 3000 h; the glass-transition temperature was determined in accordance with ASTM D3418 [24]. Two scans were performed for each BFRP type. Fig. 4 presents the DSC scans for the T_g. The first scan was useful to determine the difference in T_g between the reference and conditioned specimens. A decrease in T_g for the conditioned specimens is indicative of a plasticizing effect or chemical degradation. A higher specimen T_g after the second scan is indicative of a reversible chemical reaction and consequently not caused by any chemical degradation of the resin (such as hydrolysis). The conditioned BFRP specimens showed first-run T_g values of 101, 108, and 113 °C for Types A, B, and C, respectively. The T_g values from the

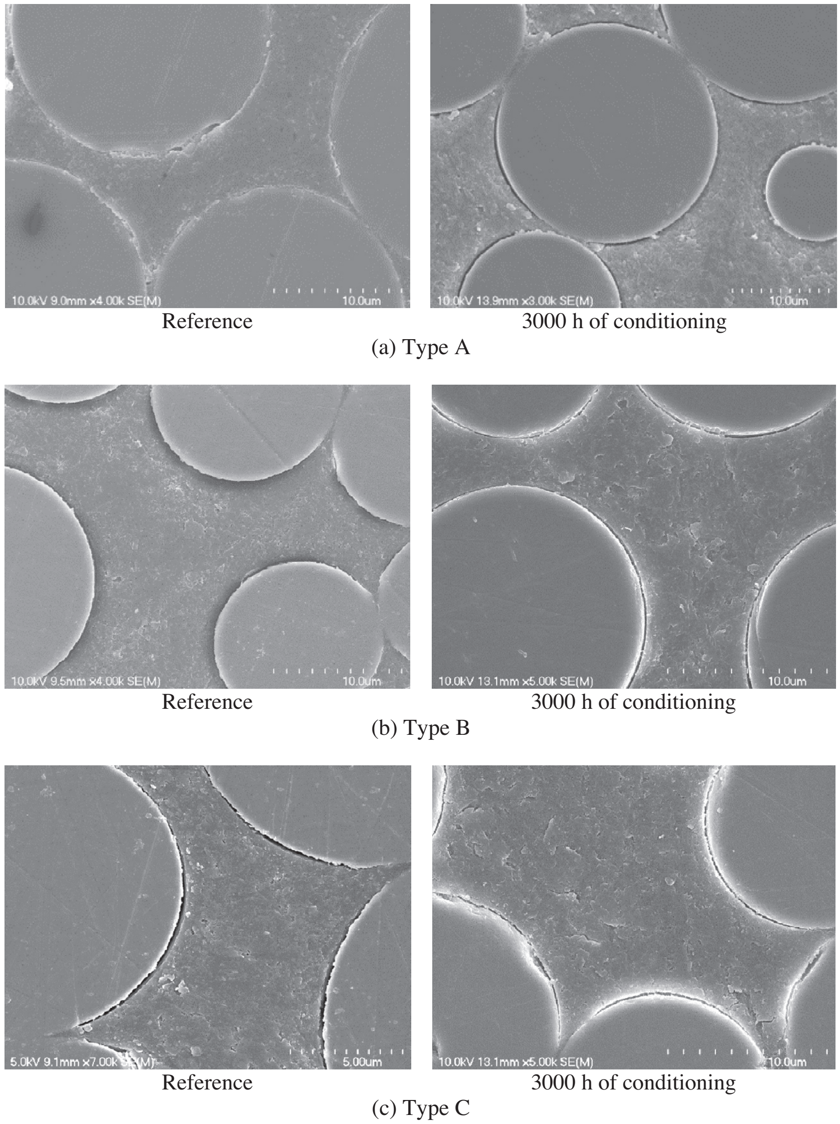


Fig. 12. Micrograph of the unconditioned and conditioned BFRP bars.

second run were 117, 121, and 134 °C for Types A, B, and C, respectively. The increased T_g in the second run can be explained by a reversible chemical reaction.

5.3.7. Moisture uptake at saturation of the conditioned BFRP bars

The moisture uptake at saturation was also determined for the specimens conditioned in the alkaline solution at 60 °C for 3000 h. The moisture-uptake ratios at saturation for the reference specimens were 0.15%, 0.56%, and 0.62% for Types A, B, and C, respectively. The moisture-uptake ratios at saturation for the conditioned specimens were 0.16%, 0.49%, and 0.59% for Types A, B, and C, respectively. Clearly, the conditioned and reference specimens had similar moisture uptakes.

5.3.8. Microstructural analysis of the reference and conditioned BFRP bars

Scanning-electron-microscopy (SEM) observations were performed to investigate microstructural changes in the BFRP bars before and after conditioning. The specimens were cut, polished, and coated with a thin layer of gold/palladium with a vapor-deposit process. The analysis was carried out on a JEOL JSM-840A microscope. The specimens were cut in one-inch lengths and placed in cylindrical molds, where epoxy resin was cast. After 24 h of curing at room temperature, the samples were removed and cut with a low-speed saw equipped with a diamond blade. Then, the specimens were polished and coated with a thin layer of gold/palladium in a vapor-deposit process. Thereafter, microstructural observations were performed with a JEOL JSM-840A microscope.

Fig. 12 presents micrographs of the reference and conditioned BFRP specimens. SEM of the reference specimens revealed some delamination or gaps between fibers and the resin (fiber–matrix interface). The micrographs show that the worst fiber–matrix interface was observed in Type C, where complete separation between fibers and resin was evident. On the other hand, the micrographs of the Type A reference specimens revealed no significant issues with the fiber–matrix interface. These results can account for the highest moisture uptake measured at saturation for the Type C bars.

The initial status of the fiber–matrix interface (reference specimens) usually gave indications of the rate of degradation and the changes in mechanical properties related specifically to the fiber–

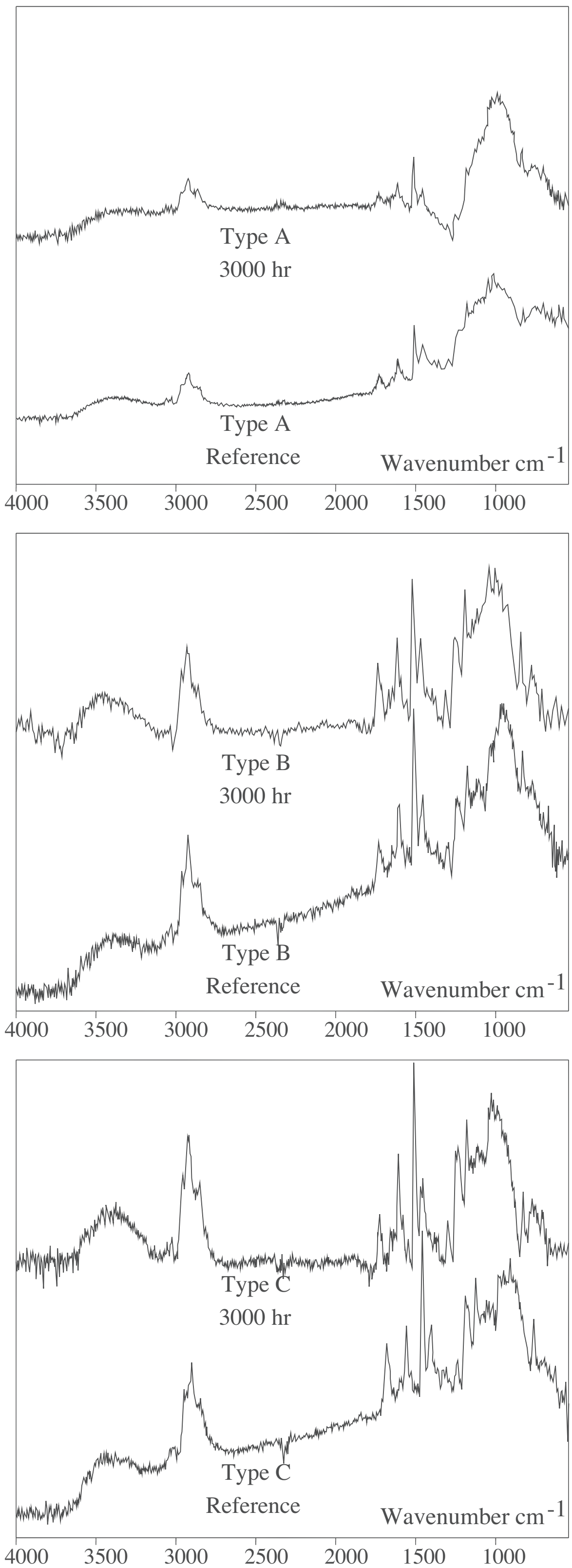


Fig. 13. FTIR spectra of the BFRP specimens.

matrix interface, such as interlaminar-shear strength. The presence of fiber–matrix delamination or voids accelerates degradation, since it allows the alkaline solution to penetrate between the fibers and resin, resulting in a higher diffusion of hydroxyl ions in the polymer matrix. Fig. 12 clearly shows the conditioning-related degradation of the fiber–matrix interface in the tested BFRP bars. Since the Type C reference specimens showed such significant delamination of the fiber–matrix interface, the degradation was noticeable in the properties related to the fiber–matrix interface. The reduction in the interlaminar-shear strength of the conditioned Type B specimens was 40.2% (59.8% retention).

5.3.9. Chemical changes in the BFRP bars after conditioning

Fourier transform infrared spectroscopy (FTIR) was used to identify any chemical change/degradation after conditioning. FTIR spectra were recorded with an ABB-Bomen (MB series) spectrometer equipped with an attenuated total reflectance device. Thirty-two scans were routinely acquired with an optical retardation of 0.25 cm to yield a resolution of 4 cm^{-1}. The reference and 3000 h conditioned specimens were investigated.

Fig. 13 shows the FTIR analysis of the reference and conditioned BFRP bars. The Type A reference and conditioned specimens evidenced very similar spectra. The hydroxyl peak did not show any significant changes, which means that the amount of hydroxyl groups present in the resins had not increased. This observation confirms that the resins did not degrade chemically when the BFRP bars were immersed in the alkaline solution at 60 °C for 3000 h. In the case of Types B and C, a very small increase in the peak related to hydroxyl groups (–OH) around 3400 cm^{-1} was observed. This could be due to a slight chemical degradation of the resin matrix through a hydrolysis reaction or more probably the presence of alkalis or water from the conditioning solution.

6. Summary and conclusions

This paper presented the result of an investigation conducted to characterize newly developed basalt fiber-reinforced polymer (BFRP) bars. The experiments included physical and mechanical testing of reference and alkali-conditioned specimens. The testing was conducted according to ACI [15] and CSA [16] as well as relevant ASTM standard test methods. Based on the test results presented herein, the following conclusions can be drawn:

1. These results confirmed that the basalt FRP (BFRP) bars meet the requirements of ACI [15] and CSA [16] with respect to physical and mechanical properties. The bond strength of the Type A BFRP bars (manufactured for prestressing purposes) obtained herein was in agreement with CEP-FIP [33] values for steel tendons. The long-term testing, however, showed significant degradation and reduction in the mechanical properties of the alkali-conditioned specimens.
2. The results confirm that the basalt fibers and resins used in this study were not affected by the conditioning. The strength degradation observed in the BFRP bars was attributed to the fiber–matrix interface (sizing), which evidenced poor bonding between the resin and basalt fibers, as confirmed by scanning electronic microscopy (SEM). Addressing this aspect is essential for producing durable BFRP bars that perform well in concrete environments.
3. The results obtained herein contribute to developing and enhancing the properties of BFRP bars. Further investigations on different BFRP products should be conducted to generate more confidence and encourage wider acceptance of this new material, which may lead to introducing the BFRP materials into the FRP design codes and standards.

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